

Operator Discovery Report: scaffold_selector_transfer_fallback

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Select a simple baseline scaffold deterministically from instance-descriptor structure to improve cross-instance transfer; uses nearest_neighbor_2opt as fallback when descriptors are uninformative.
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:clustered_local_search.
- Surviving candidate: False.
- Problem class where it helps: grid_like_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: 2eaf77a86c4666a9.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on heldout_tsplib with gap delta 0.009631.
- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.068425.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into nearest_neighbor_2opt.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Does not transplant cleanly into random_restart_2opt.
- Does not transplant cleanly into sparse_three_opt.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.
- Runtime inflation is high at 0.894435.

Transplant Results

- nearest_neighbor_2opt gap delta 0.003494, runtime delta 198.5123, distance-eval delta 8994.142857.

- cheapest_insertion_2opt gap delta 0.110815, runtime delta -4234.790657, distance-eval delta -6752063.857143.
- random_restart_2opt gap delta 0.02382, runtime delta 47.410457, distance-eval delta -6708.0.
- sparse_three_opt gap delta 0.00478, runtime delta 261.361257, distance-eval delta 5572.857143.
- clustered_local_search gap delta -0.02085, runtime delta 81.133671, distance-eval delta -4963.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.244689 vs baseline 0.235058.
- clustered_tsp operator gap 0.113212 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.082744 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.068425 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: 0.004671.
- Runtime inflation: 0.894435.
- Distance-evaluation inflation: 1.541079.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled_mean gap 0.121317 and delta vs full operator -0.005919.
- flattened_selector mean gap 0.121317 and delta vs full operator -0.005919.
- shuffled_selector mean gap 0.098866 and delta vs full operator -0.02837.